

ibm360_memory_sop_v1.json

```
{
  "id": "SOP-IBM360-002",
  "title": "IBM System/360 Model 40 Main Storage (Memory) Parity Diagnostic",
  "system": "IBM System/360 Model 40",
  "component": "2040 Main Storage Unit",
  "author": "SymbioStreams Semantic Bridge",
  "source": "IBM Field Engineering Maintenance Manual (SY22-2841)",
  "description": "Diagnostic and repair logic for memory parity errors in the Model 40 Main Storage.",
  "steps": [
    {
      "step": 1,
      "action": "Machine Check Verification",
      "details": "Check the System Control Panel for a 'Machine Check' light. Verify the Machine Check Interruption Code (MCIC) in storage locations 232-239 (decimal).",
      "checks": [
        "Storage Parity Error bit in MCIC.",
        "Storage Address Check bit in MCIC."
      ]
    },
    {
      "step": 2,
      "action": "Address Identification",
      "details": "Analyze the SAR (Storage Address Register) at the time of the check. The SAR contains the physical address where the parity error occurred.",
      "indicator": "Identify the failing bit and byte in the 4-byte (word) storage structure."
    },
    {
      "step": 3,
      "action": "Diagnostic Routine (MSD)",
      "details": "Load and run the Main Storage Diagnostics (MSD). These tests use specific bit patterns (00, FF, AA, 55) to stress the magnetic core storage.",
      "logic": "If the error is intermittent, run the 'Worst Case Pattern' loop."
    },
    {
      "step": 4,
      "action": "Core Array Isolation",
      "details": "If the error is solid, locate the specific SLT (Solid Logic Technology) card or core array board associated with the failing address.",
      "remedy": "Replace the failing SLT card or adjust the drive currents on the storage control board."
    }
  ],
  "diagnostic_flowchart": [
    {
      "condition": "Machine Check light ON, MCIC bit 4 set",
      "next": "Parity error detected. Read SAR for address."
    },
    {
      "condition": "Error occurs only on specific pattern",
      "next": "Intermittent core failure. Adjust drive currents."
    }
  ]
}
```

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.....
},
{
  "condition": "Diagnostic passes with new SLT card",
  "next": "Faulty card isolated and replaced."
}
]
}
```